get the idea of working with bucketing from our examples, you should be good to go with the other aggregations by following the documentation. Let's start with a common bucket aggregation: histograms.

13.3.1 Histograms

Histograms are neat bar charts representing grouped data. Most analytical software tools provide visual as well as data representation of histograms. Elasticsearch exposes a `histogram` bucket aggregation out of the box.

You may have worked with histograms where data is split into multiple categories based on the appropriate interval. The histograms in Elasticsearch are no different: they create a set of buckets over all the documents at a predetermined interval.

Let's take an example of categorizing books by ratings. We want to find the number of books in each rating category: 2 to 3, 3 to 4, 4 to 5, and so on. We can create a `histogram` aggregation using a set `interval` value of `1` so books fall into one-step buckets of ratings.

Listing 13.10 A `histogram` aggregation for books

```
GET books/_search
{
  "size": 0,
  "aggs": {
    "ratings_histogram": {
      "histogram": {
        "field": "amazon_rating",
        "interval": 1
      }
    }
  }
}
```

Names our aggregation

Applies the aggregation on this field

Categorizes data into buckets

Specifies the bucket interval (one unit)

The `histogram` aggregation expects the field on which we want to aggregate the buckets as well as the interval for the buckets. In this listing, we split the books based on the `amazon_rating` field with an interval of `1`. This fetches all books that fall between 3 and 4, 4 and 5, and so on. Figure 13.2 shows the response.

As you can see, running the query fetches 2 books that fall in the bucket of ratings 2 to 3, 35 books with ratings 3 to 4, and so on. The response in figure 13.2 shows that the buckets have two fields: `key` and `doc_count`. The `key` field represents the bucket classification, and the `doc_count` field indicates the number of documents that fit in the bucket.


```
"aggregations" : {
  "ratings_histogram" : {
    "buckets" : [
      {
        "key" : 3.5,
        "doc_count" : 2
      },
      {
        "key" : 4.0,
        "doc_count" : 9
      },
      {
        "key" : 4.5,
        "doc_count" : 26
      },
      {
        "key" : 5.0,
        "doc_count" : 2
      }
    ]
  }
```


Figure 13.2 Book rating aggregation as a histogram